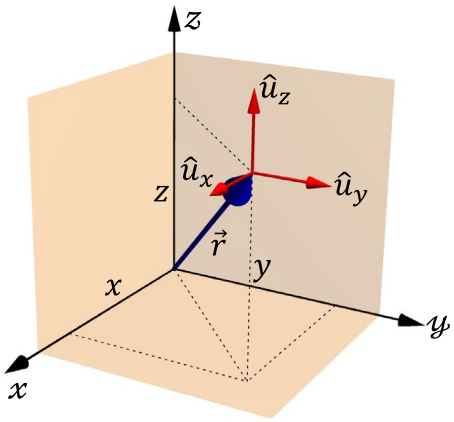
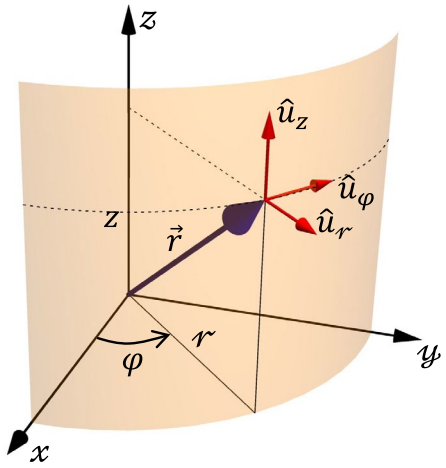
Coordenadas cartesianas



$$\vec{r} = x\,\hat{u}_x + y\,\hat{u}_y + z\,\hat{u}_z \qquad d\vec{\ell} = dx\,\hat{u}_x + dy\,\hat{u}_y + dz\,\hat{u}_z$$

$$dS_x = dy\,dz$$
$$dS_y = dx\,dz \qquad dV = dx\,dy\,dz$$
$$dS_z = dx\,dy$$

Coordenadas cilíndricas



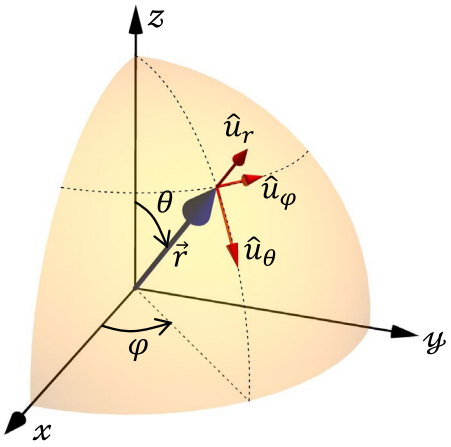
$$x = r\cos\varphi \qquad \hat{u}_x = \cos\varphi\ \hat{u}_r - \operatorname{sen}\varphi\ \hat{u}_\varphi$$
$$y = r\operatorname{sen}\varphi \qquad \hat{u}_y = \operatorname{sen}\varphi\ \hat{u}_r + \cos\varphi\ \hat{u}_\varphi$$
$$z = z \qquad \hat{u}_z = \hat{u}_z$$

$$r = \sqrt{x^2 + y^2} \qquad \hat{u}_r = \cos\varphi\ \hat{u}_x + \operatorname{sen}\varphi\ \hat{u}_y$$
$$\varphi = \tan^{-1}(y/x) \qquad \hat{u}_\varphi = -\operatorname{sen}\varphi\ \hat{u}_x + \cos\varphi\ \hat{u}_y$$
$$z = z \qquad \hat{u}_z = \hat{u}_z$$

$$\vec{r} = r\,\hat{u}_r + z\,\hat{u}_z \qquad d\vec{\ell} = dr\,\hat{u}_r + r\,d\varphi\,\hat{u}_\varphi + dz\,\hat{u}_z$$

$$dS_r = r\,d\varphi\,dz$$
$$dS_\varphi = dr\,dz \qquad dV = r\,dr\,d\varphi\,dz$$
$$dS_z = r\,dr\,d\varphi$$

Coordenadas esféricas



$$x = r\operatorname{sen}\theta\cos\varphi \qquad \hat{u}_x = \operatorname{sen}\theta\cos\varphi\ \hat{u}_r + \cos\theta\cos\varphi\,\hat{u}_\theta - \operatorname{sen}\varphi\,\hat{u}_\varphi$$
$$y = r\operatorname{sen}\theta\operatorname{sen}\varphi \qquad \hat{u}_y = \operatorname{sen}\theta\operatorname{sen}\varphi\ \hat{u}_r + \cos\theta\operatorname{sen}\varphi\ \hat{u}_\theta + \cos\varphi\,\hat{u}_\varphi$$
$$z = r\cos\theta \qquad \hat{u}_z = \cos\theta\ \hat{u}_r - \operatorname{sen}\theta\ \hat{u}_\theta$$

$$r = \sqrt{x^2 + y^2 + z^2} \qquad \hat{u}_r = \operatorname{sen}\theta\cos\varphi\ \hat{u}_x + \operatorname{sen}\theta\operatorname{sen}\varphi\ \hat{u}_y + \cos\theta\,\hat{u}_z$$
$$\theta = \tan^{-1}\left(\sqrt{x^2 + y^2}/z\right) \qquad \hat{u}_\theta = \cos\theta\cos\varphi\ \hat{u}_x + \cos\theta\operatorname{sen}\varphi\ \hat{u}_y - \operatorname{sen}\theta\,\hat{u}_z$$
$$\varphi = \tan^{-1}(y/x) \qquad \hat{u}_\varphi = -\operatorname{sen}\varphi\ \hat{u}_x + \cos\varphi\,\hat{u}_y$$

$$\vec{r} = r\,\hat{u}_r \qquad d\vec{\ell} = dr\,\hat{u}_r + r\,d\theta\,\hat{u}_\theta + r\operatorname{sen}\theta\,d\varphi\,\hat{u}_\varphi$$

$$dS_r = r^2\operatorname{sen}\theta\,d\theta\,d\varphi$$
$$dS_\theta = r\operatorname{sen}\theta\,dr\,d\varphi \qquad dV = r^2\operatorname{sen}\theta\,dr\,d\theta\,d\varphi$$
$$dS_\varphi = r\,dr\,d\theta$$

Coordenadas cartesianas

$$\vec{\nabla} f = \frac{\partial f}{\partial x}\hat{u}_x + \frac{\partial f}{\partial y}\hat{u}_y + \frac{\partial f}{\partial z}\hat{u}_z$$

$$\vec{\nabla} \cdot \vec{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$$

$$\vec{\nabla} \times \vec{F} = \left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}\right)\hat{u}_x + \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}\right)\hat{u}_y + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right)\hat{u}_z$$

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

Coordenadas cilíndricas

$$\vec{\nabla} f = \frac{\partial f}{\partial r}\hat{u}_r + \frac{1}{r}\frac{\partial f}{\partial \varphi}\hat{u}_\varphi + \frac{\partial f}{\partial z}\hat{u}_z$$

$$\vec{\nabla} \cdot \vec{F} = \frac{1}{r}\frac{\partial (r F_r)}{\partial r} + \frac{1}{r}\frac{\partial F_\varphi}{\partial \varphi} + \frac{\partial F_z}{\partial z}$$

$$\vec{\nabla} \times \vec{F} = \left(\frac{1}{r}\frac{\partial F_z}{\partial \varphi} - \frac{\partial F_\varphi}{\partial z}\right)\hat{u}_r + \left(\frac{\partial F_r}{\partial z} - \frac{\partial F_z}{\partial r}\right)\hat{u}_\varphi + \frac{1}{r}\left(\frac{\partial (r F_\varphi)}{\partial r} - \frac{\partial F_r}{\partial \varphi}\right)\hat{u}_z$$

$$\nabla^2 f = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial f}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 f}{\partial \varphi^2} + \frac{\partial^2 f}{\partial z^2}$$

Coordenadas esféricas

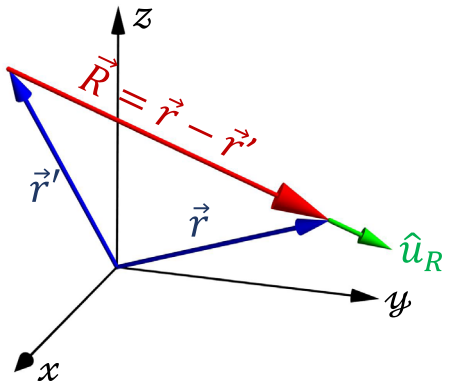
$$\vec{\nabla} f = \frac{\partial f}{\partial r}\hat{u}_r + \frac{1}{r}\frac{\partial f}{\partial \theta}\hat{u}_\theta + \frac{1}{r\,\mathrm{sen}\,\theta}\frac{\partial f}{\partial \varphi}\hat{u}_\varphi$$

$$\vec{\nabla} \cdot \vec{F} = \frac{1}{r^2}\frac{\partial (r^2 F_r)}{\partial r} + \frac{1}{r\,\mathrm{sen}\,\theta}\frac{\partial (F_\theta \,\mathrm{sen}\,\theta)}{\partial \theta} + \frac{1}{r\,\mathrm{sen}\,\theta}\frac{\partial F_\varphi}{\partial \varphi}$$

$$\vec{\nabla} \times \vec{F} = \frac{1}{r\,\mathrm{sen}\,\theta}\left(\frac{\partial (F_\varphi \,\mathrm{sen}\,\theta)}{\partial \theta} - \frac{\partial F_\theta}{\partial \varphi}\right)\hat{u}_r + \frac{1}{r}\left(\frac{1}{\mathrm{sen}\,\theta}\frac{\partial F_r}{\partial \varphi} - \frac{\partial (r F_\varphi)}{\partial r}\right)\hat{u}_\theta + \frac{1}{r}\left(\frac{\partial (r F_\theta)}{\partial r} - \frac{\partial F_r}{\partial \theta}\right)\hat{u}_\varphi$$

$$\nabla^2 f = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right) + \frac{1}{r^2\,\mathrm{sen}\,\theta}\frac{\partial}{\partial \theta}\left(\mathrm{sen}\,\theta\frac{\partial f}{\partial \theta}\right) + \frac{1}{r^2\,\mathrm{sen}^2\,\theta}\frac{\partial^2 f}{\partial \varphi^2}$$

Funciones de la posición relativa y sus derivadas



$$\vec{R} = \vec{r} - \vec{r}' = (x - x')\,\hat{u}_x + (y - y')\,\hat{u}_y + (z - z')\,\hat{u}_z$$

$$\vec{\nabla}(R^n) = -\vec{\nabla}'(R^n) = n\,R^{n-1}\hat{u}_R$$

$$\vec{\nabla} \cdot (\hat{u}_R/R^2) = -\vec{\nabla}' \cdot (\hat{u}_R/R^2) = 4\pi\,\delta^3(\vec{R})$$

$$\vec{\nabla} \cdot (R^n\,\hat{u}_R) = -\vec{\nabla}' \cdot (R^n\,\hat{u}_R) = (n+2)R^{n-1} \quad (n \neq -2)$$

$$\vec{\nabla} \times (R^n\,\hat{u}_R) = -\vec{\nabla}' \times (R^n\,\hat{u}_R) = 0$$

$$\nabla^2(1/R) = \nabla'^2(1/R) = -4\pi\,\delta^3(\vec{R})$$

Doble producto de vectores

$$\vec{A}\cdot(\vec{B}\times\vec{C})=\vec{B}\cdot(\vec{C}\times\vec{A})=\vec{C}\cdot(\vec{A}\times\vec{B})$$

$$\vec{A}\times(\vec{B}\times\vec{C})=(\vec{A}\cdot\vec{C})\vec{B}-(\vec{A}\cdot\vec{B})\vec{C}$$

Derivadas vectoriales

$$\vec{\nabla}(f\,g)=f\,\vec{\nabla}g+g\,\vec{\nabla}f$$

$$\vec{\nabla}\cdot(\vec{\nabla}f)=\nabla^2 f=\frac{\partial^2 f}{\partial x^2}+\frac{\partial^2 f}{\partial y^2}+\frac{\partial^2 f}{\partial z^2}$$

$$\vec{\nabla}(\vec{A}\cdot\vec{B})=\vec{A}\times(\vec{\nabla}\times\vec{B})+\vec{B}\times(\vec{\nabla}\times\vec{A})+(\vec{A}\cdot\vec{\nabla})\,\vec{B}+(\vec{B}\cdot\vec{\nabla})\,\vec{A}$$

$$\vec{\nabla}\cdot(\vec{\nabla}\times\vec{A})=0$$

$$\vec{\nabla}\cdot(f\,\vec{A})=f\,\vec{\nabla}\cdot\vec{A}+\vec{A}\cdot\vec{\nabla}f$$

$$\vec{\nabla}\times(\vec{\nabla}f)=0$$

$$\vec{\nabla}\cdot(\vec{A}\times\vec{B})=\vec{B}\cdot(\vec{\nabla}\times\vec{A})-\vec{A}\cdot(\vec{\nabla}\times\vec{B})$$

$$\vec{\nabla}\times(\vec{\nabla}\times\vec{A})=\vec{\nabla}\,(\vec{\nabla}\cdot\vec{A})-\nabla^2\vec{A}$$

$$\vec{\nabla}\times(f\,\vec{A})=\vec{\nabla}f\times\vec{A}+f\,(\vec{\nabla}\times\vec{A})$$

$$\vec{\nabla}\times(\vec{A}\times\vec{B})=(\vec{\nabla}\cdot\vec{B})\,\vec{A}-(\vec{\nabla}\cdot\vec{A})\,\vec{B}+(\vec{B}\cdot\vec{\nabla})\vec{A}-(\vec{A}\cdot\vec{\nabla})\vec{B}$$

Teoremas integrales

$$\int_{\vec{a}}^{\vec{b}}\vec{\nabla}f\cdot d\vec{\ell}=f(\vec{b})-f(\vec{a})$$ Teorema del gradiente

$$\int_V\vec{\nabla}\cdot\vec{A}\;dv=\oint_S\vec{A}\cdot d\vec{s}$$ Teorema de la divergencia (Gauss)

$$\int_S(\vec{\nabla}\times\vec{A})\cdot d\vec{s}\;=\oint_L\vec{A}\cdot d\vec{\ell}$$ Teorema del rotacional (Stokes)

Integrales que aparecen con frecuencia

$$\int\frac{dx}{(x^2+a^2)^{1/2}}=\ln\left(x+\sqrt{x^2+a^2}\right)$$

$$\int\frac{x\,dx}{(x^2+a^2)^{1/2}}=\sqrt{x^2+a^2}$$

$$\int\frac{dx}{x^2+a^2}=\frac{1}{a}\arctan(x/a)$$

$$\int\frac{x\,dx}{x^2+a^2}=\frac{1}{2}\ln(x^2+a^2)$$

$$\int\frac{dx}{(x^2+a^2)^{3/2}}=\frac{x}{a^2\sqrt{x^2+a^2}}$$

$$\int\frac{x\,dx}{(x^2+a^2)^{3/2}}=-\frac{1}{\sqrt{x^2+a^2}}$$

Función δ-Dirac

$$\delta(x)=\begin{cases}0, x\neq 0\\ \infty, x=0\end{cases}$$

$$\int_{-\infty}^{\infty}\delta(x)\,dx=1$$

$$\int_{-\infty}^{\infty}f(x)\,\delta(x-a)\,dx=f(a)$$

$$\delta^3(\vec{r})=\begin{cases}0, \vec{r}\neq\vec{0}\\ \infty, \vec{r}=\vec{0}\end{cases}$$

$$\int_{V_\infty}\delta^3(\vec{r})\,dv=1$$

$$\int_{V_\infty}f(\vec{r})\,\delta^3(\vec{r}-\vec{a})\,dv=f(\vec{a})$$

Polinomios de Legendre

$$\frac{1}{|\vec{r}-\vec{r}'|}=\frac{1}{r}\sum_{n=0}^{\infty}\left(\frac{r'}{r}\right)^n P_n(\cos\theta')$$

$$P_0(x)=1$$
$$P_1(x)=x$$
$$P_2(x)=\tfrac{1}{2}(3x^2-1)$$
$$P_3(x)=\tfrac{1}{2}(5x^3-3x)$$